

Bryce Kan

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EDUCATION

University of Southern California

Aug. 2024 – Dec. 2026

Masters of Science in Computer Science

Coursework: Algorithms, Machine Learning, Natural Language Processing, Trustworthy ML, Hardware for ML

University of Washington

Aug. 2018 – Dec. 2021

Bachelor of Science in Molecular and Cell Biology

TECHNICAL SKILLS

Languages: Python, C++/C, CUDA, Java, SQL

Frameworks/Libraries: PyTorch, OpenCV, sentence-transformers, Scikit-learn, Hugging Face, Pandas, Matplotlib, Numpy

Developer Tools: Git, Linux, Jupyter, VS Code, SLURM, SSH, Conda

EXPERIENCE

SWE-Agentic AI Master's Intern

May. 2026 – August. 2026

Honeywell

Atlanta, GA

- buildl mcp server
- chatbot sql queries
- aws, model deployment

Graduate Research Assistant

Mar. 2025 – Present

USC Machine Learning Center

Los Angeles, CA

RaMem: Contextual Reinstatement for Long-Term Agentic Memory

- Built **RAG-based agentic memory pipeline** enabling LLM agents to recall evidence across multi-session conversations.
- Engineered **temporal evidence filtering** with session metadata to remove semantically related but invalid memories.
- Implemented hybrid retrieval with **Reciprocal Rank Fusion (RRF)** over semantic search and BM25, improving F1 by **11.6 points** across GPT-4o, GPT-4.1-mini, and Qwen3-8B.
- Reduced retrieved-memory context by **50%** while preserving performance against state-of-the-art agent memory baselines.

Student-Job Recommendation System

- Led end-to-end development as **1st author**, mentoring **2 Master's students** and collaborating with PhD candidates
- Tackled **cold-start scenarios** using synthetic explanation generation via **GPT-4o-mini distillation** into **Llama-3-8B-Instruct** for student-teacher learning
- Implemented **multi-step fine-tuning** with **dual optimization framework**, simultaneously training for ranking (**InfoNCE contrastive loss**)
- Achieved **up to 2.6× (Recall@5)** and **over 3× (NDCG@5)** improvement over LLM baselines

Research Associate

Jun. 2023 – Jul. 2024

Bruker

San Jose, CA

- Developed custom **Python pipelines** to automate experimental data analysis, adapting to evolving scientific demands
- Designed interactive **Plotly dashboards** to visualize complex biological datasets, enabling faster data-driven insights

Process Engineer

Jan. 2022 – Jun. 2023

Applied Materials

Santa Clara, CA

- Launched Lab Inventory Database as final product, delivering full workflow traceability and reducing processing time by **80%**
- Built automated data visualization pipelines using Pandas and Plotly to extract insights from **terabyte-sized datasets**

PROJECTS

Multimodal Cancer Prediction with Whole-Slide Histopathology | *PyTorch, OpenCV, Segmentation*

- Integrated gigapixel WSI and clinical-text features through patch-level segmentation and attention-based clustering analysis.

Efficient Fine-Tuning of lightweight BERT Models with LoRA [Link] | *PEFT, Transformers, Pytorch*

- Applied parameter-efficient fine-tuning to BERT-Tiny and BERT-2L, achieving **~70% accuracy improvement**

MLP Optimization via Kernel Fusion and Vectorization [Link] | *CUDA, cuBLAS, GPU Optimization*

- Implemented a cuBLAS-based 2-layer MLP with fused kernels, reducing kernel launches by **33%** and achieving **1.14× speedup**

PUBLICATIONS

Yang, W, **Kan, B.** et al.(2026). *RaMem: Contextual Reinstatement for Long-Term Agentic Memory*. Neurips 2026. [Link]

Kan, B. et al. (2026). *De-conflating Preference and Qualification: Constrained Dual-Perspective Reasoning for Job Recommendation with Large Language Models*. arXiv. [Link]

Seo, D.-O., Zhang, E. T., Piantadosi, S. C., Marcus, D. J., Motard, L. E., **Kan, B. K.**, et al. (2021). *A locus coeruleus to dentate gyrus noradrenergic circuit modulates aversive contextual processing*. Neuron. [Link]